

Benjamín Rodríguez Romo

MSC IN BIOENGINEERING | BIOENGINEERING AND COMPUTER ENGINEERING DEGREES

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I am an Engineer with a multidisciplinary background in bioengineering and computer engineering, with research experience in biomedical imaging, machine learning, computational biomechanics, statistical analysis, and scientific software development. Experienced in developing Python-based pipelines, 3D image and model analysis workflows, finite-element simulations, and interactive visualization tools, while also maintaining hands-on experience in experimental biomechanics, biomaterials, and laboratory methods.

PROFESSIONAL EXPERIENCE - RESEARCH

EMPA - Swiss federal laboratories for materials science and technology.

Dübendorf, Switzerland

Research Intern | Biomedical Engineering & Structural Mechanics Group

February 2026 – January 2027

- Applied statistical analysis and machine-learning methods to biomechanical and experimental datasets for quantitative characterization and predictive modeling
- Developed and evaluated finite-element models in Abaqus/Explicit for biomechanical applications.
- Developed Python-based pipelines for automated simulation execution, monitoring, failure detection, ODB data extraction, and quantitative post-processing.
- Designed interactive dashboards for simulation setup, monitoring, and visualization of kinematic, mechanical, energetic, and geometric results.
- Integrated computational, statistical, and visualization workflows to support model comparison, parameter studies, and interpretation of biomechanical experiments.

RUSH UNIVERSITY MEDICAL CENTER

Chicago, Illinois, USA

Exchange Research Student | Biomechanics Department

June 2025 – July 2025

- Developed Python workflows for alignment, correspondence, and quantitative analysis of 3D anatomical models.
- Applied statistical shape models to generate and study representative anatomical samples for morphology-based analyses.
- Estimated and compared cartilage thickness and anatomical morphology using 3D models.
- Developed automated pipelines for extracting geometric anatomical measurements from 3D models.
- Developed customized 3D Slicer extensions integrating image processing, anatomical analysis, visualization, and study-specific computational workflows.

BIOENGINEERING CENTER, UAI.

Viña Del Mar, Chile

Research Assistant | B3MAT Research Group

March 2022 - December 2025

- Performed mechanical testing and quantitative characterization of biomaterials and additively manufactured structures.
- Processed and analyzed micro-CT, MRI, and X-ray data for biomedical imaging and 3D reconstruction applications.
- Designed and manufactured experimental samples and prototypes using CAD and additive manufacturing techniques.
- Supported multidisciplinary R&D projects involving biomaterials, biomechanics, medical imaging, and computational analysis.
- Managed biomechanical and biomaterials laboratory equipment, experimental workflows, and laboratory operations.
- Supervised undergraduate students conducting research projects and experimental work.

PROFESSIONAL EXPERIENCE - TEACHING

ADOLFO IBÁÑEZ UNIVERSITY

August 2022 - December 2024

Teaching assistant of the Faculty of Engineering and Sciences

Supported course delivery through laboratory preparation and supervision, problem-solving sessions, assessment development, grading, and evaluation of student projects.

- **Undergraduate courses:** Bioinformatics and Data Science, Fundamentals of Data Science, Biomechanics, Biomaterials, Bioprocesses, Introduction to Engineering, and Innovation & Entrepreneurship.
- **Postgraduate courses:** Experimental Techniques in Biomaterials.

EDUCATION

MSc in Bioengineering , Universidad Adolfo Ibáñez (GPA: 6.6/7.0; Thesis: 7.0/7.0)	2023-2026
Professional Degree in Bioengineering , Universidad Adolfo Ibáñez	2019–2026
Professional Degree in Computer Engineering , Universidad Adolfo Ibáñez	2019–2026

Skills - Manufacturing and Laboratory

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- **Programming & Data Analysis:** Python, SQL, R, C, Git, LaTeX, Excel.
 - **Machine Learning & Scientific Computing:** PyTorch, SciPy, scikit-image, statistical analysis, machine-learning workflows, biomedical data processing.
 - **Biomedical Imaging & 3D Analysis:** CT/MRI segmentation, 3D reconstruction, visualization, geometric analysis, and model manipulation using 3D Slicer, VTK, Open3D, ImageJ, and Cell Profiler.
 - **Computational Mechanics:** Finite-element modeling and analysis using Abaqus/Explicit for quasi-static and dynamic simulations, including biomechanical applications.
 - **Mechanical Testing & Experimental Characterization:** ASTM-based tensile, compression, and bending testing; microscopy; porosity, pore-size, and density characterization.
 - **CAD & Manufacturing:** Fusion, Inventor, FDM 3D printing, laser cutting, and polymer filament extrusion.
 - **Biomaterials & Laboratory Methods:** Biomaterial synthesis and preparation, including hydroxyapatite and natural calcium carbonate, sample preparation, and laboratory experimentation.
 - **Cloud & Development Tools:** Microsoft Azure and interactive scientific dashboard development using Trame and Streamlit.

Publications, Conference and Proceedings

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- IEEE 15th International Conference on Pattern Recognition Systems (ICPRS 2025). Proceedings and oral presentation. CAMalyzer: A 3DSlicer extension for AI segmentation and generation of proximal femur 3D models. 1st author. (doi: [10.1109/ICPRS66293.2025.11302867](https://doi.org/10.1109/ICPRS66293.2025.11302867))
 - Orthopedic Research Society 2025 Annual Meeting (ORS). Conference poster. *Pipeline for MRI automatic segmentation and generation of a 3D model of the femoral head using Machine Learning*. 1st author.
 - Orthopedic Research Society 2026 Annual Meeting (ORS). Conference Poster. *A Framework for Quantifying Tibial Cartilage Deformation After Activity In Vivo via MRI*. 2nd author.
 - IEEE Journal of Biomedical and Health Informatics. Journal publication. *CAMalyzer: An Automated MRI-Based 3D Slicer Framework for Femoral Morphology Analysis in Femoroacetabular Impingement Syndrome*. First author (Under review)
 - ACS Biomaterials Science & Engineering. *Microstructural heterogeneity of β -TCP granules regulates early stem cell adhesion and spatial organization, revealed by quantitative morphometric analysis*. Third author. (Under review)

Scholarships and awards

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| • Academic Honor Scholarship Adolfo Ibáñez University.
<i>For outstanding students with a GPA above 6.0 of 7.0.</i> | 2019 - 2023 |
| • PSU Score Scholarship, Adolfo Ibáñez University.
<i>Equivalent to the SAT exam. Outstanding score: 760 of 850</i> | 2018 |

Skills - Languages

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- English (B2)
 - Spanish (Native speaker)

Skills - Extraprogramatic and Student Activities

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| • COIL experience for Biomimetics class with Bucknell University. | 2025 |
| • Member of a student organization recognized by the University (Biohuella). | 2022-2023 |
| • Basketball Team of Universidad Adolfo Ibáñez. | 2019-2022 |